Department of Computing
The Hong Kong Polytechnic University

COMP4442 Service and Cloud Computing
Lab 2: File Management with AWS S3

0. What we will learn

- How to manage your files with AWS S3
- How to share your files as an URL

1. Install AWS CLI

Windows: https://awscli.amazonaws.com/AWSCLIV2.msi

macOS: https://awscli.amazonaws.com/AWSCLIV2.pkg

AWS CLI stands for the general common line interface (CLI) of Amazon Web Services. With AWS CLI, you can manage many AWS services such as EC2, S3, and RDS. In this lab, we only consider file management with S3. We need to install the install AWS CLI first. You can download the installation file from internet and install it on your computer. (For windows user, download the `*.msi` file; for macOS user, download the `*.pkg` file).

You can use the following command to see whether AWS CLI is successfully installed.

```
AWS --version
```

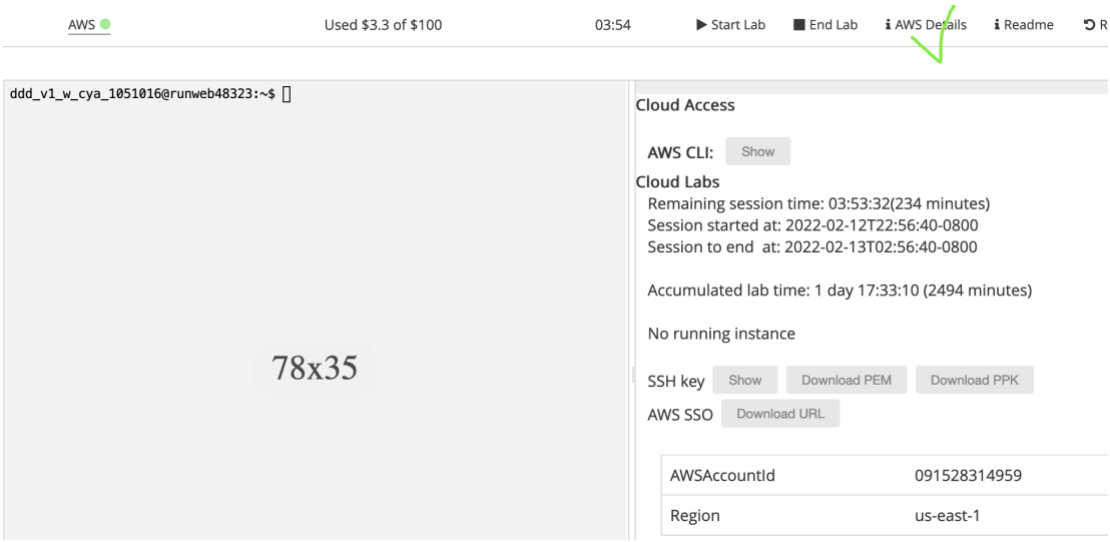
If successfully installed, you will see the versions of AWS CLI. An example is shown in the following figure.

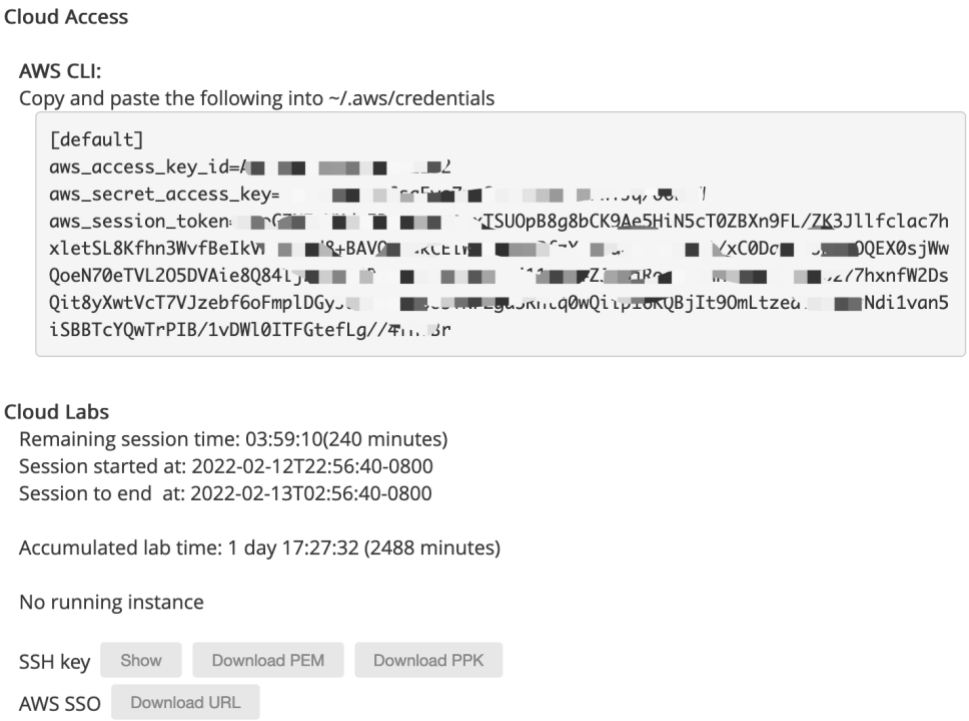
```
C:\Users\ZHANG>aws --version
aws-cli/2.1.27 Python/3.7.9 Windows/10 exe/AMD64 prompt/off

C:\Users\ZHANG>
```

2. Create a Bucket

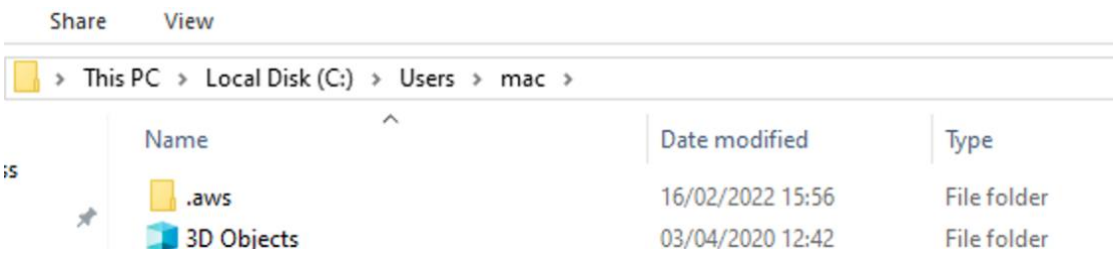
Before starting the lab, don't forget to store the **latest credentials** to the `~/.aws/credentials` file. (The token will be invalid after three hours)

An example of credential is as follows:

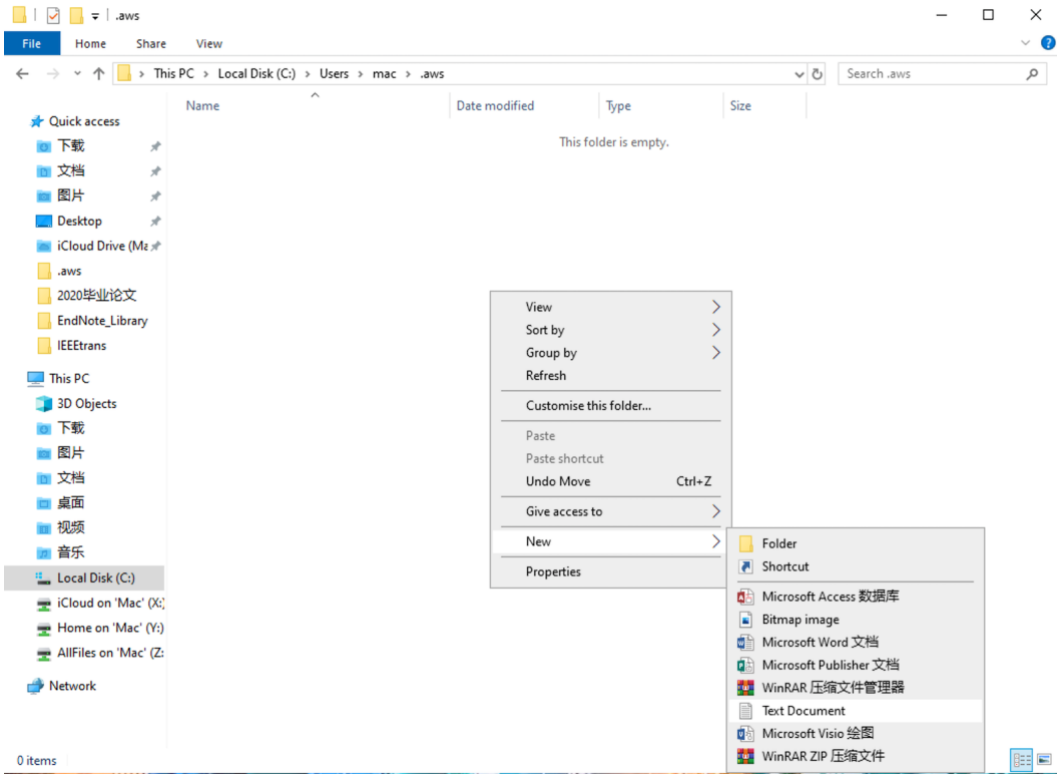




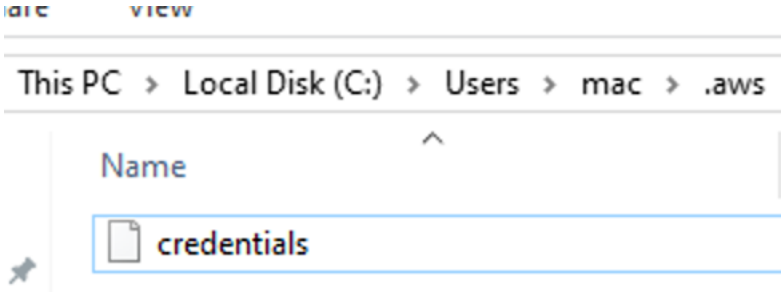
For windows user, you can go to `C:\Users\<user_name>\.aws` folder. If the folder does not exists, please create it.



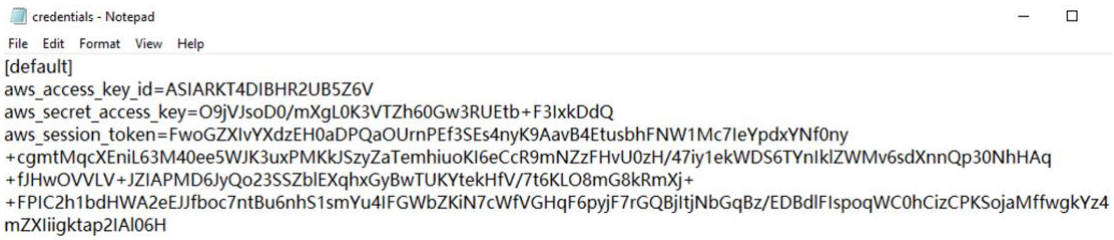
Go to .aws folder, create an empty file



Rename the filename to be credentials (remember there is no extension name).



Use notepad to open it and copy the content of credentials to it,

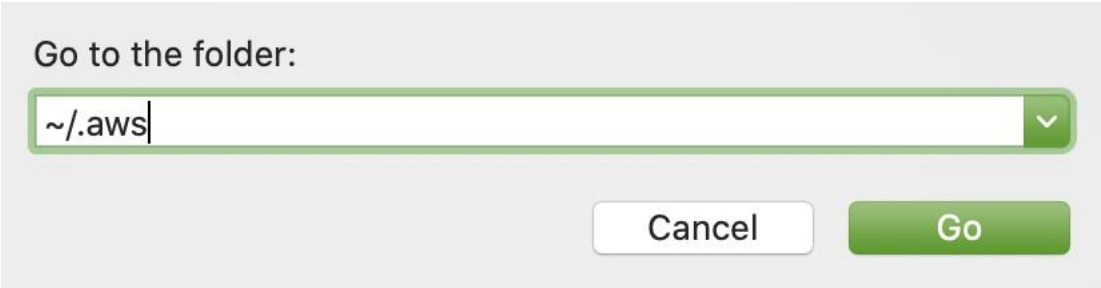


Save the file.

For MacOS user, you can go to `~/.aws/`, if there is no such folder, you need to create it first.

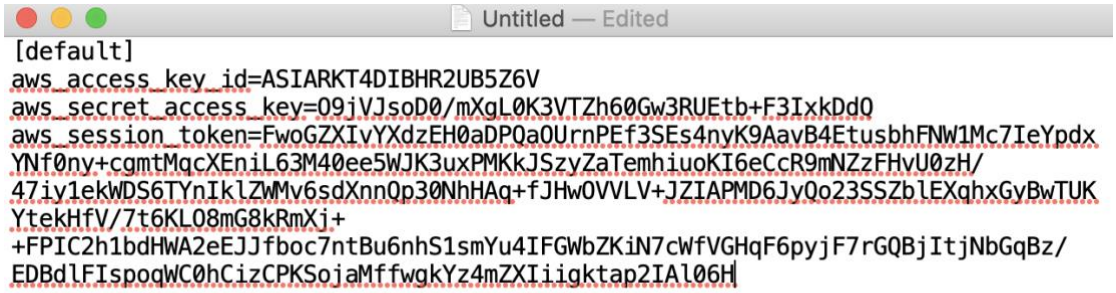
Use Go function -> Go to Folder

Input the path '~/.aws' and click Go.

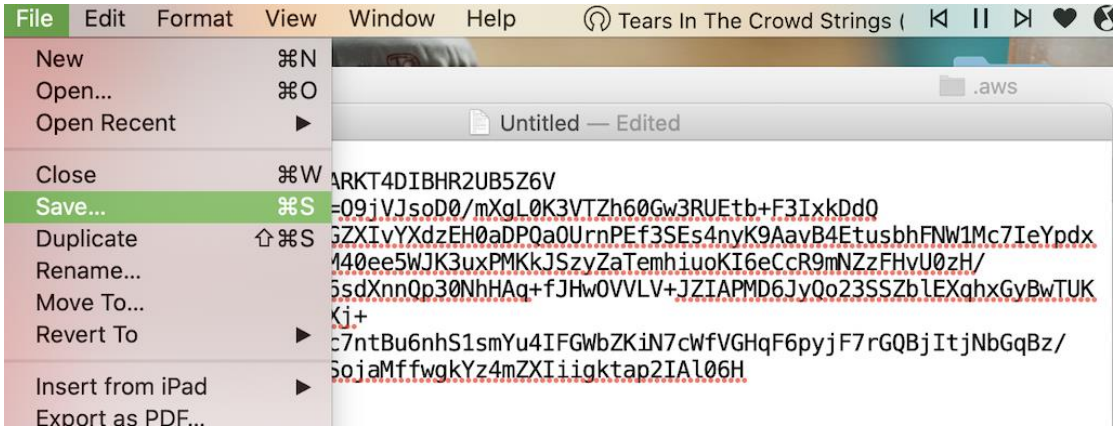


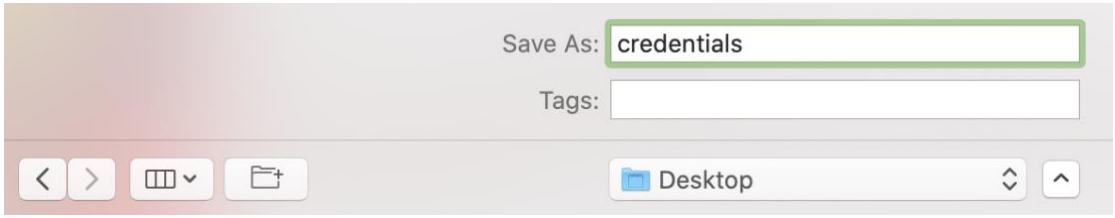
We open the TextEdit.app of macOS, and then press 'shift + command + T' to change the mode of TextEdit.app to be pure text input mode,

And then we copy the content of credentials to the editor,

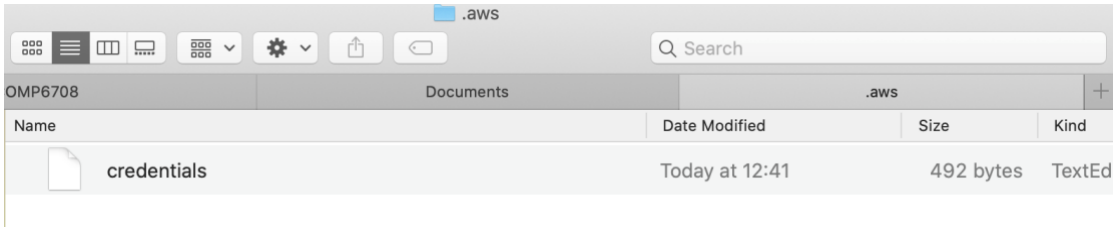


Save this file with the name of ‘credentials’ to ~/.aws





We save the credentials file to the desktop first, and then copy it to ~/.aws folder.



After creating the credentials file, you can use the following command to create a bucket:

```
aws s3 mb s3://yyy
```

in which “mb” means “make bucket”, and “yyy” is the name of the bucket.

For example, I can create a bucket named comp4442-lab2-docs and the command is as follows:

```
C:\Users\ZHANG>aws s3 mb s3://comp4442-lab2-docs
make_bucket: comp4442-lab2-docs

C:\Users\ZHANG>aws s3 ls
2021-03-05 20:22:57 comp4442-lab2-docs
2021-03-05 11:31:13 comp4442-lab2-mingjin
2021-02-26 11:12:23 elasticbeanstalk-us-east-1-762251576369

C:\Users\ZHANG>
```

After creating a bucket, you can manage files just like in a unix-based system. That is, you can use the commands such as “ls” to list files, and “rm” to remove files.

For example, I can copy my local file `aws-logo.png` to the bucket comp4442-docs that we just created. The command is as follows:

```
aws s3 cp aws-logo.png s3://comp4442-lab2-docs/
```

The following figure shows the command.

```
D:\>cd "Me\Courses\cloud computing\Lab-02"

D:\Me\Courses\cloud computing\Lab-02>aws s3 cp aws-logo.png s3://comp4442-lab2-docs/
upload: .\aws-logo.png to s3://comp4442-lab2-docs/aws-logo.png

D:\Me\Courses\cloud computing\Lab-02>aws s3 ls s3://comp4442-lab2-docs/
2021-03-05 20:25:17     168985 aws-logo.png

D:\Me\Courses\cloud computing\Lab-02>
```

3. Exercise

- Try other commands, such as “aws s3 sync”, and “aws s3 rb”.
- For more information about the AWS CLI and commands, refer to

  https://docs.aws.amazon.com/cli/latest/userguide/cli-chap-welcome.html

  https://awscli.amazonaws.com/v2/documentation/api/latest/reference/s3/index.html